

Prajwal Tidke

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Professional Summary

Data Engineer with 4+ years of experience designing **ETL pipelines**, **cloud-based data platforms**, and **BI dashboards**. Hands-on experience applying **machine learning and data science** to real-world problems, including knowledge-grounded chatbots, biosignal prediction, anomaly detection, and accident risk analysis. Skilled in **transformers**, **temporal models**, **generative augmentation**, and **interpretable ML**.

Experience

Research Assistant (Part-Time), San Diego State University – San Diego, CA Feb 2025 – Present

- Fine-tuned **LLaMA** with **KBLaM** to build a multilingual, privacy-aware mental health chatbot, grounded in **psychological knowledge triples**.
- Trained **MLP**, **LSTM**, and **LagMLP** with **temporal attention** on biosignals (PPG, ECG, EDA), achieving **25% MAE reduction**.
- Applied **WGAN** augmentation and **Monte Carlo Dropout** to quantify uncertainty and validated generalization with **LOSO-CV**.

Associate Data Engineer, TresVista Analytics – Pune, India Oct 2023 – Aug 2024

- Directed a **2-person analytics team** supporting **CPPIB (Canada Pension Plan Investment Board)**, delivering automated ETL workflows and reporting dashboards.
- Engineered **Databricks (PySpark)** + **Apache Airflow** pipelines to consolidate data from **heterogeneous sources**, reducing ingestion time by **50%**.
- Automated financial model refreshes and reporting with **Python**, **VBA**, **Power Automate**, reducing manual effort by **70%**.
- Established **SLAs**, **validation rules**, and **reporting standards** with stakeholders, ensuring reports met timeliness and accuracy requirements.

Senior Data Engineer, LTIMindtree – Pune, India Jul 2020 – Oct 2023

- Led a **data engineering team of 2** as Project Lead, managing deliverables, delivery governance, and client coordination.
- Built and optimized **SAP BODS ETL pipelines** integrating SAP ECC, HANA, SQL Server, and Oracle into a BI warehouse, accelerating delivery by **80%**.
- Automated recurring ETL processes, saving **1.5 FTEs annually** and improving operational reliability.
- Migrated legacy SAP pipelines to **AWS Glue** and designed Oracle-to-Snowflake transfers, achieving **~ 50% cost savings**.
- Administered **SAP BODS environments** (job servers, user roles, patching, compliance, upgrades) under enterprise governance frameworks.

Education

San Diego State University, USA, MS in Data Science (GPA: 4.0/4.0) Aug 2024 – May 2026

IIITDM Jabalpur, India, BTech in Computer Science (GPA: 3.2/4.0) Aug 2016 – May 2020

Projects

Anomaly Detection with WGAN Jan 2025 - May 2025

- Conducted a research study to evaluate whether **WGAN-based data augmentation** can enhance the performance of **traditional and modern anomaly detection methods** on imbalanced tabular datasets.
- Demonstrated **20–30% performance improvement** under severe data imbalance using proposed WGAN based approach.

TrafficSensAI: Predictive Accident Analysis Aug 2024 - Dec 2024

- Developed an ML workflow with **DBSCAN clustering** and **Random Forest** to predict accident severity and detect hotspots.
- Built interactive **ArcGIS visualizations** to surface accident trends and support **urban safety planning**.

Technical Skills

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers (LLaMA, KBLaM), LSTM, WGAN, GANs, XGBoost, Random Forest, Transfer Learning, Ensemble Methods, Attention Mechanisms

Programming & Scripting: Python, SQL, R, Java, C/C++ , Bash, VBA

Data Engineering & ETL: Databricks (PySpark), Apache Airflow, SAP BODS, AWS Glue

Cloud Platforms & Databases: AWS (S3, Redshift, EC2, Athena, DynamoDB, SageMaker, Rekognition), Azure (AI Foundry, Machine Learning Studio, Cognitive Services), PostgreSQL, Oracle, Snowflake, Firebase, MongoDB

Visualization & Reporting: Tableau, Power BI, ArcGIS, Qlik, SAP BI

Tools & Others: Git, Notion, JIRA, ServiceNow, Microsoft Excel, DBeaver